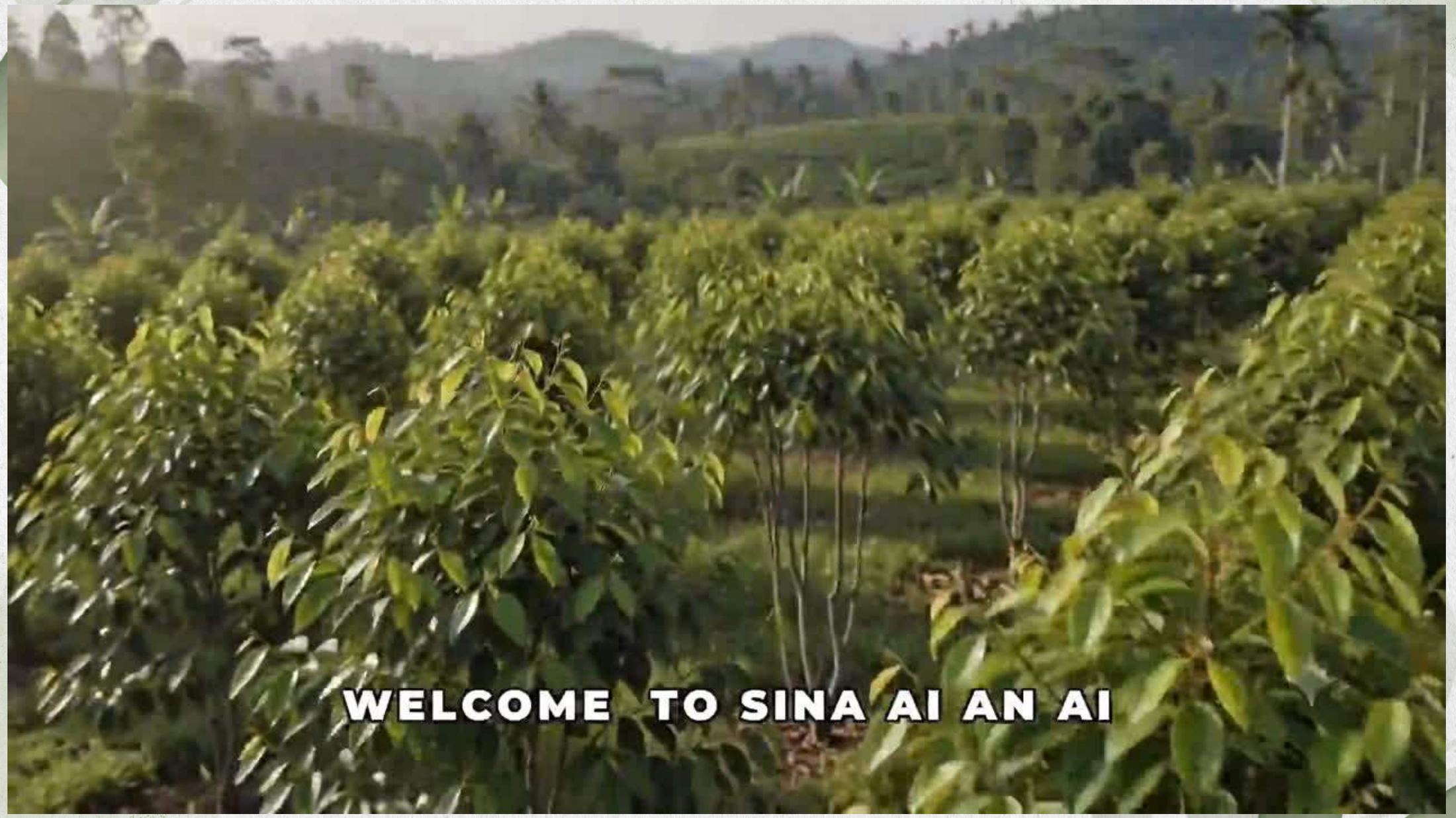
# AI- DRIVEN SMART CINNAMON GROWTH MONITORING & AUTOMATED HARVESTING SYSTEM

**R26-IT-102**


WELCOME TO SINA AI AN AI

# TEAM

Cinna AI



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# PROBLEM & MOTIVATION

## Context

Cinnamon is one of Sri Lanka's most valuable export crops. However, several cultivation and harvesting activities still rely heavily on manual labour, experience and visual judgement.

## Observed Problems

- Cinnamon harvesting requires highly skilled manual labour.
- Bark readiness and stem condition are often judged manually.
- Disease and insect damage can be difficult to identify at an early stage.
- Manual peeling can cause inconsistent bark quality and product wastage.
- Labour shortages affect harvesting efficiency and production capacity.
- Existing processes provide limited real-time monitoring and automated decision support.

# SOLUTION OVERVIEW

Cinna AI is an integrated AI, IoT and robotic system designed to monitor cinnamon growth, identify plant health conditions and automate precision cinnamon bark harvesting.

## Key Features

- Smart cinnamon growth and environmental monitoring
- AI-based disease and insect damage detection
- Harvest readiness prediction
- Cinnamon bark boundary detection using computer vision
- Automatic harvesting coordinate generation
- Intelligent robotic peeling control
- Bark length and width measurement
- Real-time machine monitoring and control
- Safety controls including Start, Stop and Emergency Stop

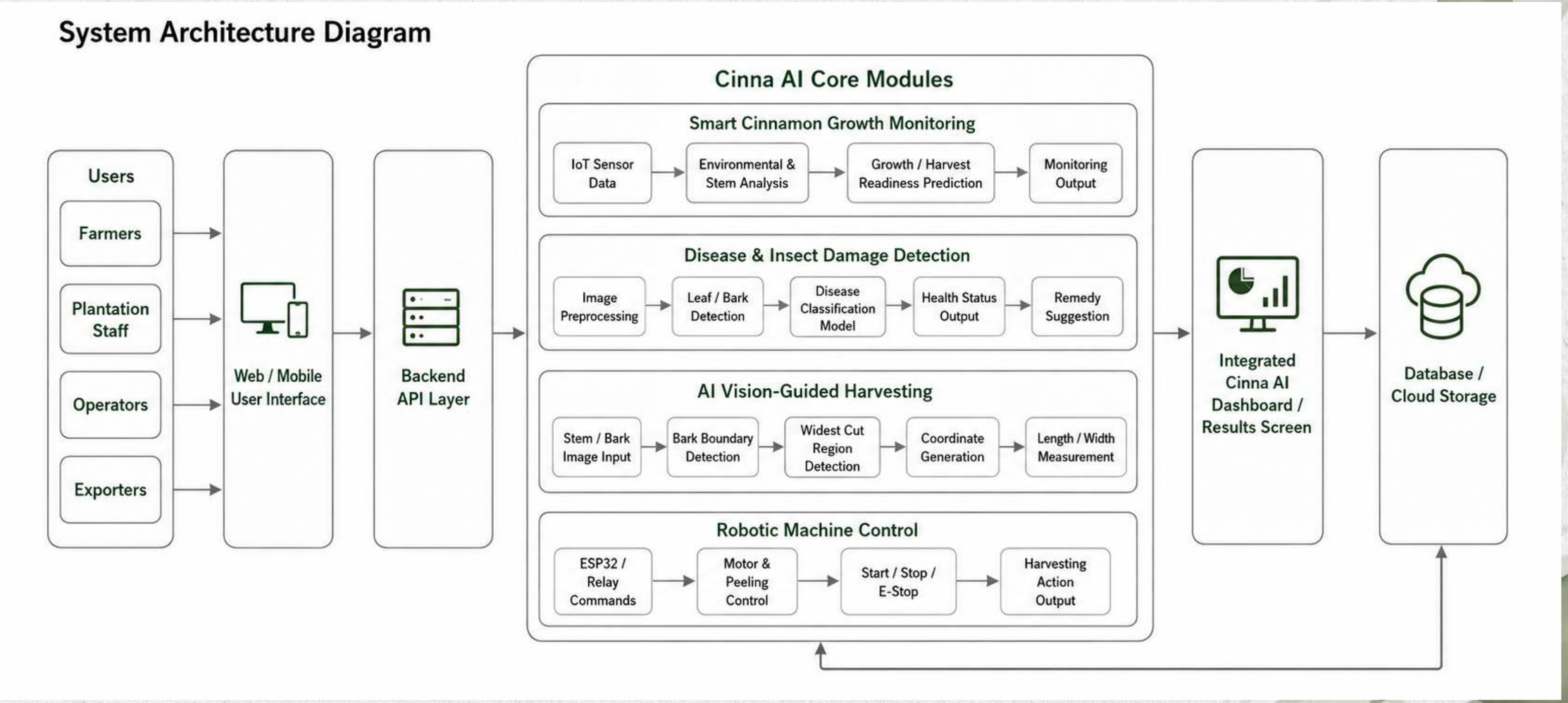
## Non-Functional Goals

- High detection accuracy
- Fast AI processing
- Easy-to-use web interface
- Safe robotic operation
- Reliable real-time communication

# OVERALL RESEARCH GAP

| Feature | Plantix App | Agrio App | Referring to Research [1] | Referring to Research [2] | Suggested Solution (Cinna AI) |
|---|---|---|---|---|---|
| Cinnamon-specific intelligent monitoring system | ✗ | ✗ | ✗ | ✗ | ✓ |
| Disease & insect damage detection through image capture | ✓ | ✓ | ✗ | ✗ | ✓ |
| Growth and harvest readiness prediction | ✗ | ✗ | ✓ | ✗ | ✓ |
| IoT-based environmental and stem monitoring | ✗ | ✗ | ✗ | ✓ | ✓ |
| Cinnamon bark boundary detection using AI vision | ✗ | ✗ | ✗ | ✗ | ✓ |
| Automatic harvesting coordinate generation | ✗ | ✗ | ✗ | ✗ | ✓ |
| AI-guided robotic bark harvesting / peeling | ✗ | ✗ | ✗ | ✓ | ✓ |
| One integrated AI + IoT + Robotics platform | ✗ | ✗ | ✗ | ✗ | ✓ |

# SYSTEM ARCHITECTURE & INTEGRATION



# SYSTEM ARCHITECTURE & INTEGRATION

**Components**

Component 01- Smart Cinnamon Growth Monitoring & Harvest Readiness Prediction- IT22092960
Component 02- Cinnamon Disease and Insect Damage Detection- IT22092656
Component 03- AI Vision-Guided Robotic Cinnamon Bark Harvesting- IT22065926

**Integration Points**

- REST API communication
- JSON-based data exchange
- IoT sensor data transmission
- Image uploads for AI analysis
- AI-generated harvesting coordinates
- ESP32 communication
- Relay-based motor control
- Shared database
- Integrated Cinna AI dashboard

# COMMERCIALIZATION

**Target Users / Market** – Cinnamon farmers, plantations, processing centers and exporters in Sri Lanka.

**Value Proposition** – AI-driven smart cinnamon monitoring and robotic harvesting system that improves accuracy, reduces manual labour and supports better harvesting decisions.

**Cost & Feasibility** – Built using open-source AI, affordable IoT sensors, ESP32 controllers and cloud technologies, making it practical for commercial deployment.

**Adoption Plan** – Prototype testing → Plantation field trials → Performance improvement → Industry partnerships → Commercial deployment.

# THANK YOU

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